

SHRI KRISHNASWAMY COLLEGE FOR WOMEN

**TN SKILLS PROJECT
[SEM 3]**

**EDUGENIE: GOOGLE GEMINI POWERED
LEARNING ASSISTANT**

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EduGenie – Google Gemini Powered Learning Assistant

Project Description

EduGenie is a lightweight AI-powered educational assistant that simplifies learning through generative AI. Designed for students of different academic levels, it enables users to ask questions, understand complex concepts through simplified explanations, generate quizzes, receive learning recommendations, and summarize educational passages.

The project uses FastAPI for the backend and a simple HTML/CSS/JavaScript interface. Cloud-based Google Gemini services provide Q&A, quiz generation, summarization, and learning recommendations, while the project also includes a local LaMini-Flan-T5 model for concept explanation.

Scenarios

- A student wants to know about oceans and rivers and uses EduGenie to ask “Which is the largest ocean?”
- A student wants to check her understanding of “The Pythagoras Theorem” and uses the Generate Quiz feature.
- A learner exploring SQL requests a structured learning path from beginner to advanced topics with guidance and resources.

Project Objective

The objective is to bring several learning-support functions into one simple web interface so that students can receive AI-assisted answers, explanations, quizzes, summaries, and learning guidance without switching between separate tools.

Pre-requisites

Python 3.10+

The project documentation specifies Python 3.10 or later. Python is used for the FastAPI backend and AI modules.

- * Download Python 3.10+ for the operating system.
- *Run the installer and enable “Add Python to PATH”.
- *Complete the installation prompts.
- *Verify installation with `python --version` in a terminal.

FastAPI Framework

FastAPI is used to create the backend API and route requests to the different EduGenie modules.

HTML & CSS

HTML and CSS are used for the web interface. The project uses the `/templates` and `/static` folders.

Google Gemini API Key

A Google Gemini API key is required for cloud-based AI functions. The key should be generated through Google AI Studio and stored securely in the environment configuration.

- *Sign in to Google AI Studio.
- *Create or select the required project.
- *Generate a Gemini API key.
- *Store the key in the `.env` file and do not publish it to GitHub.

Uvicorn and Jinja2

Uvicorn is the ASGI server used to run the FastAPI application locally. Jinja2 is used as the HTML templating engine.

Installation examples: `pip install uvicorn` | `pip install jinja2`

Requirements and Project Architecture

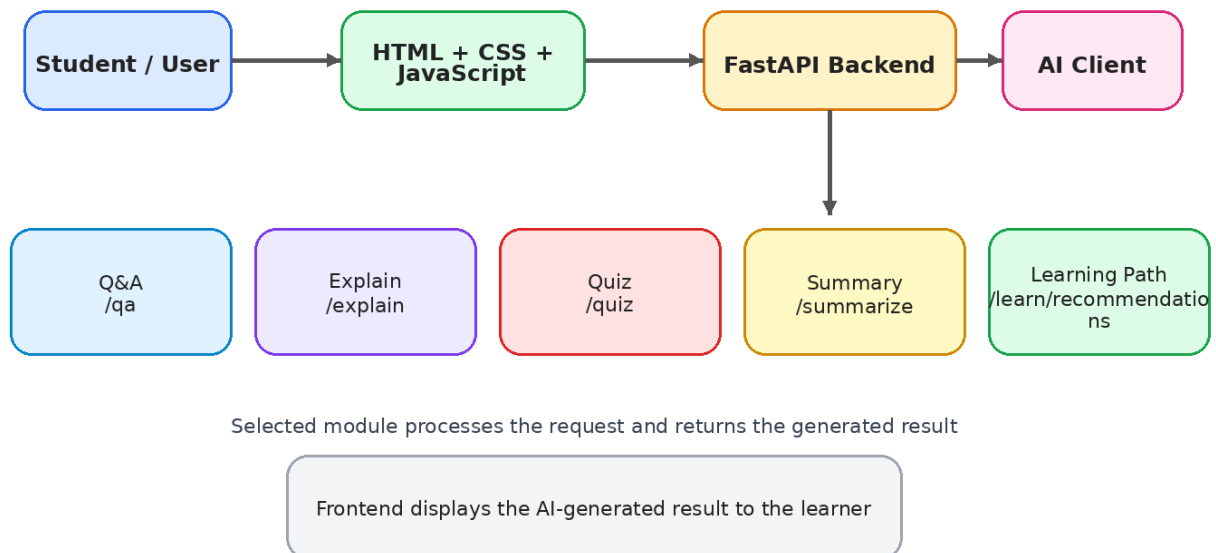
Software Requirements

- Operating System: Windows 10/11, macOS, or Linux.
- Python 3.10 or later.
- FastAPI and Uvicorn.
- Google Gemini API access.
- HTML, CSS and JavaScript capable web browser.
- Jinja2 templating engine.
- Transformers, PyTorch, sentencepiece and accelerate for the local explanation model.
- Visual Studio Cod vve or another modern code editor.

Hardware Requirements

- Processor: modern Intel/AMD processor or equivalent.
- RAM: 8 GB minimum; 16 GB is recommended when running the local AI model.
- Storage: sufficient free space for Python packages and model files.
- Stable internet connection for Google Gemini API calls.

EDUGENIE - SYSTEM ARCHITECTURE



EduGenie – Colorful system architecture and workflow

Project Workflow

EDUGENIE - PROJECT MILESTONES

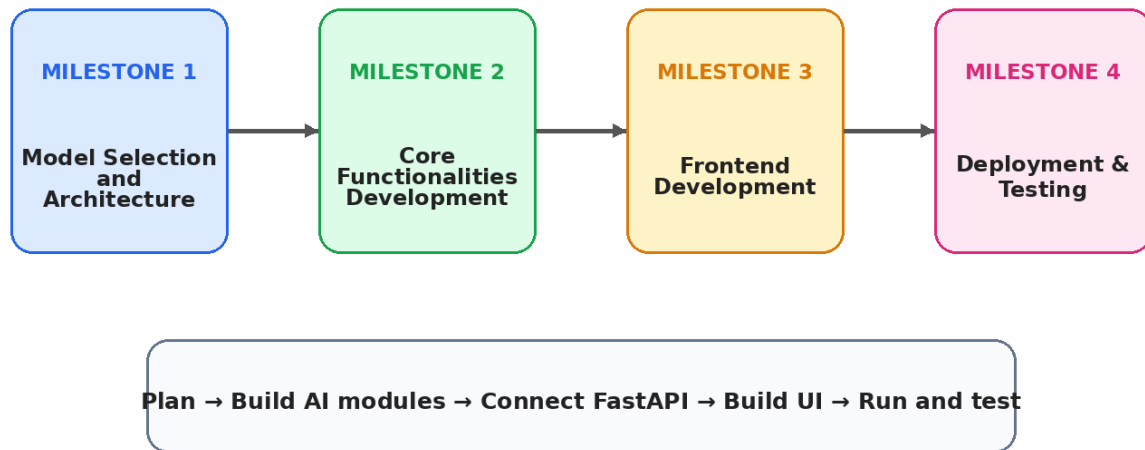


Fig. 3.1 EduGenie – Project milestones

The project workflow is organized into four major milestones: model selection and architecture, core functionality development, frontend development, and deployment/testing.

Workflow Summary

- *Select the cloud and local AI models required by each feature.
- *Implement the individual EduGenie modules.
- *Expose each module through a FastAPI endpoint.
- *Connect the web interface to the backend using POST requests.
- *Run the application locally with Uvicorn and perform functional testing.

MILESTONE 1: MODEL SELECTION AND ARCHITECTURE

Activity 1.1: Select AI Models

AI Model	Used For	Role in EduGenie
Google Gemini	Q&A, quiz, summarization, learning path	Cloud-based generative AI
LaMini-Flan-T5-783M	Concept explanation	Lightweight local explanation model

Folder Architecture

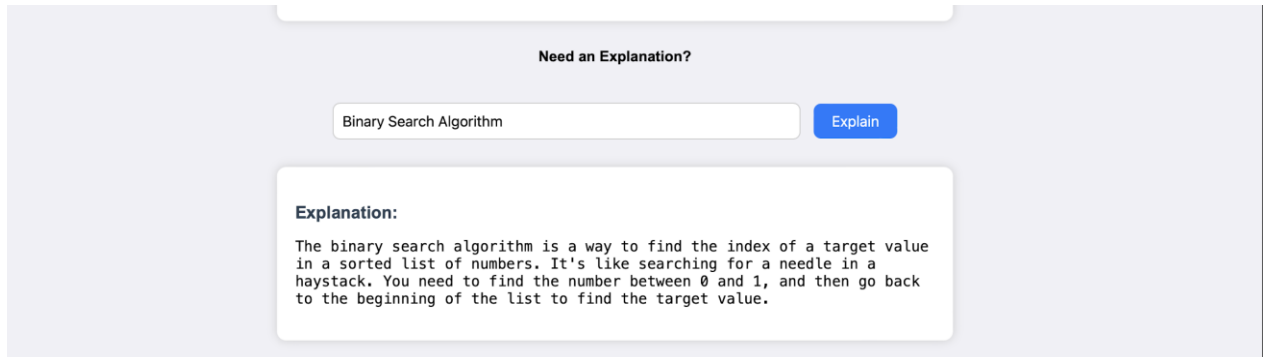
File / Folder	Purpose
main.py	FastAPI application
ai_client.py	Gemini client and model configuration
explanation_module.py	Concept explanation logic
qna.py	Question answering
quiz_module.py	Quiz generation
summary_module.py	Summarization
learning_path.py	Learning recommendations
templates/index.html	HTML frontend
static/style.css	Frontend styling
requirements.txt	Python dependencies
test_api.py	API testing

MILESTONE 2: CORE FUNCTIONALITIES DEVELOPMENT

Activity 2.1: Module Implementation

Explanation Module

EduGenie uses the LaMini-Flan-T5 model for concept explanation. The module is intended to break complex topics into simple, readable explanations suitable for beginners and self-learners.

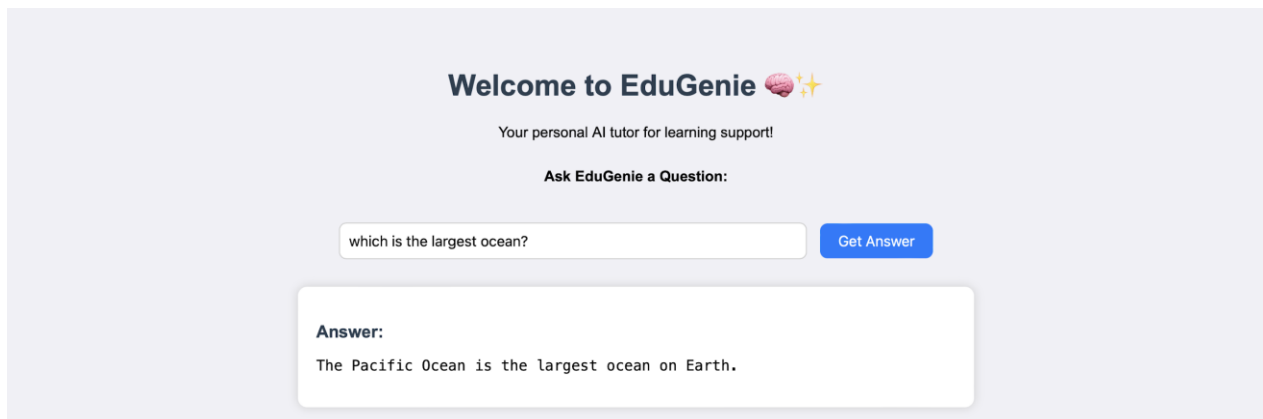


The screenshot shows a web interface for the Explanation Module. At the top, it says "Need an Explanation?". Below this is a text input field containing "Binary Search Algorithm" and a blue "Explain" button. Underneath the input field is a white box with the heading "Explanation:" and a paragraph of text: "The binary search algorithm is a way to find the index of a target value in a sorted list of numbers. It's like searching for a needle in a haystack. You need to find the number between 0 and 1, and then go back to the beginning of the list to find the target value."

Explanation module – application view

Q&A Module

The Q&A module uses Google Gemini to answer general knowledge and academic questions. The user enters a question and receives an AI-generated response through the /qa endpoint.



The screenshot shows a web interface for the Q&A Module. At the top, it says "Welcome to EduGenie" with a brain icon and two stars. Below this is the text "Your personal AI tutor for learning support!". Underneath is the heading "Ask EduGenie a Question:". Below this is a text input field containing "which is the largest ocean?" and a blue "Get Answer" button. Underneath the input field is a white box with the heading "Answer:" and a paragraph of text: "The Pacific Ocean is the largest ocean on Earth."

Q&A module – application view

QUIZ AND SUMMARIZATION MODULES

Quiz Module

EduGenie generates three multiple-choice questions from a given topic or passage, with four options for each question. The quiz response is structured so that the frontend can display the questions and evaluate the selected answer.

Generate a Quiz:

Pythagoras theorem

Generate Quiz

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☐ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

Check Answer

Quiz generation – question view

Quiz:

Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☒ The relationship between the sides of a right-angled triangle.
- ☐ The relationship between the area and perimeter of a triangle.
- ☐ The relationship between the sides of any triangle.

Check Answer

✓ Correct!

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

- ☐ $a + b = c$

Quiz generation – answer feedback view

Summary Module

The summarization feature uses Gemini to convert longer educational text into a concise, easy-to-understand summary while retaining the important information.

Summarize a Paragraph:

The Industrial Revolution, which began in the late 18th century, mark

Summarize

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

Summary module – application view

LEARNING PATH MODULE

Learning Recommendations

The learning_path module uses Google Gemini to generate a structured learning path for a requested topic. The project source describes progression from beginner to advanced concepts and the inclusion of useful learning resources.

Get Learning Recommendations:

Get Recommendations

Learning Recommendations for "SQL":

```
## SQL Learning Path: From Zero to Hero

This learning path is structured to progressively introduce SQL
concepts, starting from the basics and gradually advancing to more
complex topics. It's designed to be adaptive: feel free to adjust the
pace and delve deeper into areas that particularly interest you.

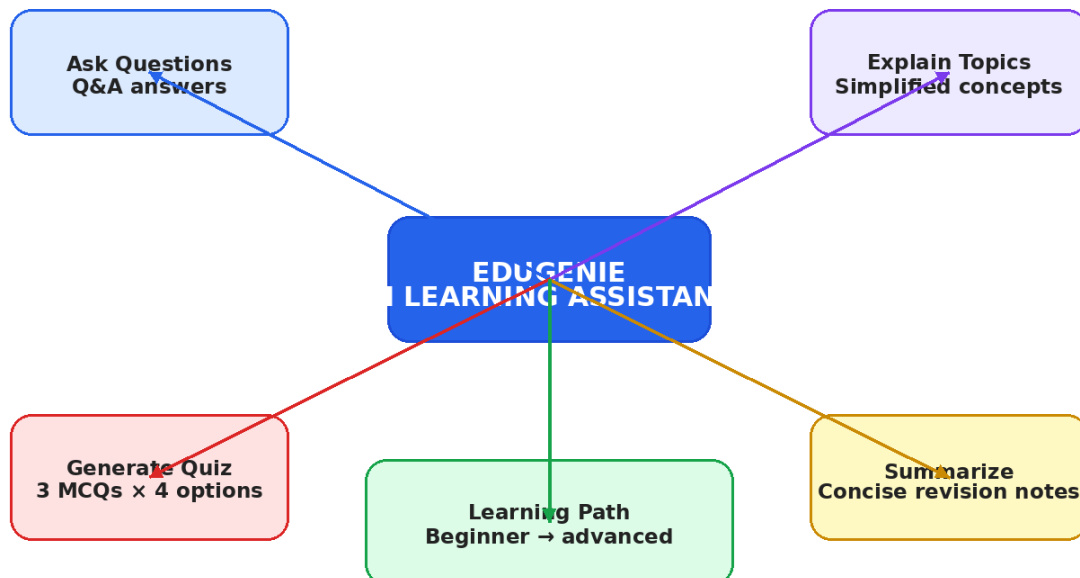
**I. Beginner Level: Building a Foundation**

**(Estimated Time: 1-2 weeks)**

* **Key Topics:**
  * What is a Database and SQL? (Relational Model, DBMS)
  * Basic Syntax (SELECT, FROM, WHERE)
  * Data Types (INT, VARCHAR, DATE, etc.)
  * Filtering Data (WHERE clause with comparison operators, logical
operators: AND, OR, NOT)
  * Ordering Results (ORDER BY)
  * Limiting Results (LIMIT/OFFSET or TOP)
  * Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)
```

Learning recommendations – application view

EDUGENIE - FEATURE MIND MAP



EduGenie – Feature mind map

Module Flow

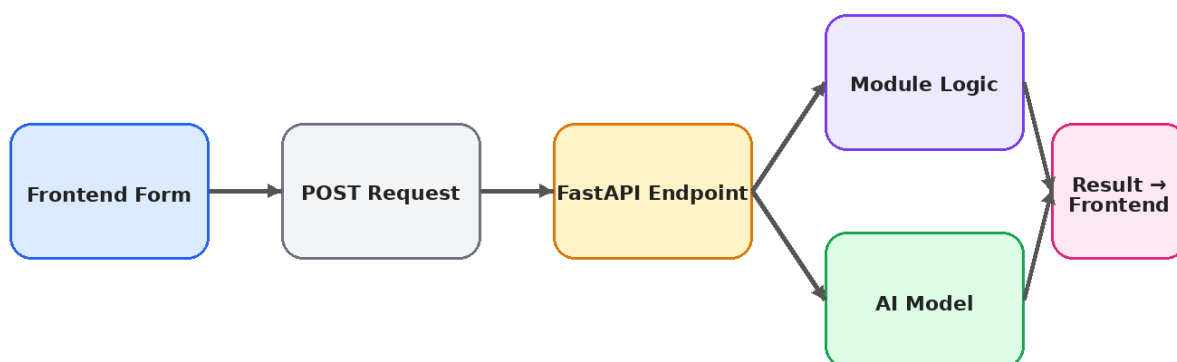
Explanation → </explain> | Q&A → </qa> | Quiz → </quiz> | Summary → </summarize> | Learning Path → </learn/recommendations>

ACTIVITY 2.2: BACKEND API WITH FASTAPI

The FastAPI backend defines RESTful endpoints for each EduGenie module and connects each endpoint to its respective module logic.

Endpoint	Module / Purpose
/qa	Question answering
/explain	Concept explanation
/quiz	Quiz generation
/summarize	Summarization
/learn/recommendations	Learning recommendations
/health	Server health check

FASTAPI - MODULE AND ENDPOINT FLOW



Endpoints: /qa • /explain • /quiz • /summarize • /learn/recommendations

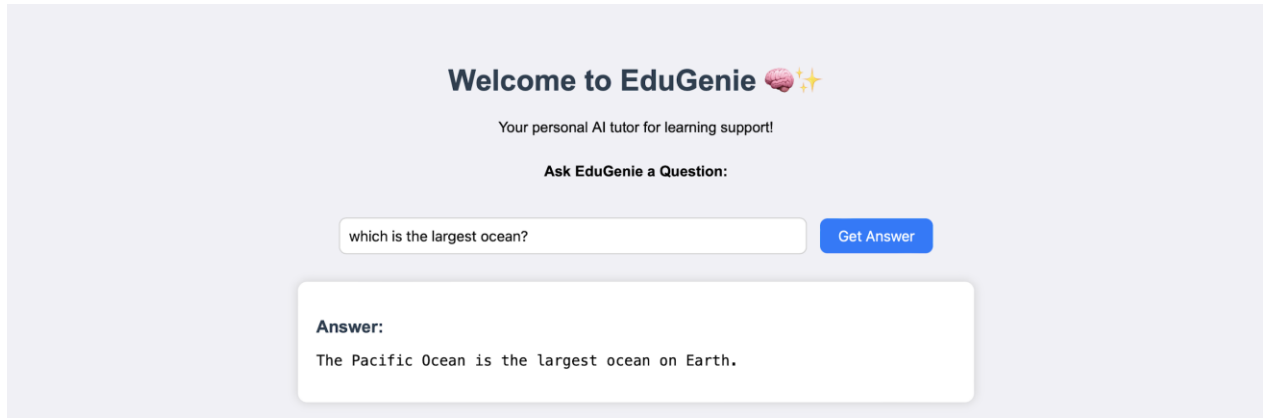
FastAPI module and endpoint flow

API Flow: Frontend form submission → POST request → FastAPI endpoint → module logic → AI model → generated result → frontend display.

MILESTONE 3: FRONTEND DEVELOPMENT

Activity 3.1: Build Web Interface

- HTML interface in templates/index.html provides the learning functions and input areas.
- CSS styling in static/style.css controls the layout, buttons, input areas and result containers.
- JavaScript sends requests to the FastAPI endpoints and places the generated results on the page.



EduGenie – integrated feature interface

Activity 3.2: Live Integration

Form submission sends a POST request to the FastAPI backend. The selected module processes the input and the generated result is displayed in the web interface.

MILESTONE 4: DEPLOYMENT AND TESTING

Activity 4.1: Run Locally

The application can be run locally using Uvicorn.

Command: `python -m uvicorn main:app --reload`

Navigate to: `http://127.0.0.1:8000`

Activity 4.2: Functional Testing

- Ask a question and verify that an answer is returned.
- Enter a topic and verify that an explanation is produced.
- Generate a quiz and check the multiple-choice options.
- Submit educational text and verify that a concise summary is returned.
- Request a topic learning path and verify that recommendations are displayed.
- Use `/health` to confirm that the backend is running.

EXPLORING EDUGENIE

Asking Questions

Welcome to EduGenie 🧠✨

Your personal AI tutor for learning support!

Ask EduGenie a Question:

Get Answer

Answer:

The Pacific Ocean is the largest ocean on Earth.

Asking questions

Explanation of Any Topic

Need an Explanation?

Explain

Explanation:

The binary search algorithm is a way to find the index of a target value in a sorted list of numbers. It's like searching for a needle in a haystack. You need to find the number between 0 and 1, and then go back to the beginning of the list to find the target value.

Explanation of a topic

Summarising Long Paragraphs

Summarize a Paragraph:

Summarize

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

Summarising long paragraphs

GENERATING QUIZZES AND LEARNING RECOMMENDATIONS

Generating Quizzes

EduGenie generates three questions with four options each. The application provides answer checking and feedback after the learner selects an option.

Generate a Quiz:

Pythagoras theorem Generate Quiz

Quiz:
Q1: What does the Pythagorean theorem describe?

- ☐ The relationship between the angles of a triangle.
- ☐ The relationship between the sides of a right-angled triangle.
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Check Answer

Generating quizzes

Quiz:
Q1: What does the Pythagorean theorem describe?

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Check Answer

✓ Correct!

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- ☐ $a + b = c$

Quiz answer feedback

Learning Recommendations

EduGenie guides learners from beginner level to advanced level and can provide learning resources and stepwise guidance for the requested topic.

Get Learning Recommendations:

SQL

Get Recommendations

Learning Recommendations for "SQL":

SQL Learning Path: From Zero to Hero

This learning path is structured to progressively introduce SQL concepts, starting from the basics and gradually advancing to more complex topics. It's designed to be adaptive: feel free to adjust the pace and delve deeper into areas that particularly interest you.

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- * Ordering Results (ORDER BY)
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- * Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)

Learning recommendations

CONCLUSION

EduGenie has been developed as an accessible AI-powered educational assistant that combines cloud-based intelligence with a simple web interface. It supports question answering, simplified explanations, quiz generation, summarization, and structured learning recommendations.

The project demonstrates practical integration of generative AI with a FastAPI backend and a browser-based frontend. Its modular structure separates the main learning functions, making the system easier to understand, test and extend.

The development process includes challenges such as maintaining the quality of AI-generated responses, handling model/API configuration, integrating different modules, and presenting generated results clearly. These challenges provide practical learning in prompt design, API integration, frontend-backend communication and AI application development.

Future Enhancements

- Voice-based interaction for hands-free learning.
- Multilingual support.
- Mobile application and offline capabilities.
- Progress tracking dashboards and learning metrics.
- Gamified learning elements such as badges and streaks.
- Adaptive learning paths based on learner activity.
- Group study and teacher/parent dashboards.
- Integration with Learning Management Systems such as Moodle or Google Classroom.
- Image and PDF input for doubt solving and summarization.
- Real-time synchronization and smart notification.

